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Review article

Emerging biomarkers of delayed graft function in kidney transplantation

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ABSTRACT

Delayed Graft Function (DGF) is one of the most common early complications in kidney transplantation, associated with poor graft outcomes, prolonged post-operative hospitalization and higher rejection rates. Given the severe shortage of high-quality organs for transplantation, DGF incidence is expected to raise in the next years because of the use of nonstandard kidneys from Extended Criteria Donors (ECD) and from Donors after Circulatory Death (DCD). Alongside conventional methods for the evaluation of renal allograft [*e.g.* serum creatinine Glomerular Filtration Rate (GFR), needle biopsy], recent advancements in omics technologies, including proteomics, metabolomics and transcriptomics, may allow to discover novel biomarkers associated with DGF occurrence, in order to identify early preclinical signs of renal dysfunction and to improve the quality of graft management. Here, we gather contributions from basic scientists and clinical researchers to describe new omics studies in renal transplantation, reporting the emerging biomarkers of DGF that may implement and improve conventional approaches.

1. DGF definition, incidence and impact

Kidney transplantation is the treatment of choice for patients with end-stage renal failure as it provides a prolonged survival and a better quality of life than dialysis. Complications during the first post-transplant year can affect graft function, representing a great social and economic burden for the healthcare system; in particular, post-transplant graft function can range from total anuria or non-oliguric Acute Tubular Necrosis (ATN) to Early Graft Function (EGF), Slow Graft Function (SGF), Delayed Graft Function (DGF) or immediate function [1]. Definitions of DGF are mainly based on serum creatinine values or dialysis requirement (*e.g.* the need for dialysis in the first 10 days after transplant, the return to maintenance hemodialysis within the first 4 days after transplantation, the serum creatinine increased or remained unchanged or decreased <10%/day during 3 consecutive days after the transplant); differences in DGF definition may influence the incidence of DGF reported worldwide, ranging approximately between 4 and 10% in living donor transplants [2] and 19%–70% in deceased donor kidney transplantations [3]. Among these, the most accepted definition of DGF is the need of at least one dialysis within the first week oftransplant [1,4,5]; based on this definition, in a retrospective cohort study, Helferet al. reported an incidence of 69.3% on 517 deceased donor kidney transplant recipients, with a significant lower graft survival in patients who experienced DGF longer than two weeks [4], while Melih et al. reported an incidence rate of 43.5% among 62 patients who received a deceased donor kidney transplant [3]. Unfortunately, the postoperative dialysis requirement is not a standardized procedure as it may depend on clinicians' clinical practice [6]; conversely, the definition of functional DGF (fDGF), based on the failure of serum creatinine to decrease by at least 10% daily on 3 consecutive days during the first week post-transplant irrespective of dialysis requirement, better reflects the functional graft recovery after IR injury [7].

When prolonged, DGF may cause readmission within the first 30 post-transplant days, acute rejection and, in the worst case, graft failure; in the long term, indeed, patients with DGF presented an overall 10% lower graft survival rate, compared to patients with EGF, and approximately 1.5 times higher risk of graft loss at 5 years [8]. In this scenario, a predictive model for DGF is a major objective of the worldwide

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transplant community as it could help clinicians in early identification and the correct management of recipients at high risk of developing DGF after transplantation, allowing prompt pharmacological interventions [6]. Irish and colleagues suggested a predictive model of the patient's risk for developing DGF and graft failure, based on more than 15 recipient-, donor- and organ-related variables, reporting that a more than 50% probability of DGF was associated with a 2-fold higher risk of graft failure [6]; conversely, Zaza and colleagues suggested a predictive model based on few and inexpensive clinical variables of the recipients, showing a lower incidence of DGF with the following parameters: young age, low body weight and short-term dialysis before transplantation [8].

Concerning the economic viewpoint, including travel expenses and drug treatments during dialysis sessions, kidney transplant saves costs in long-term compared to maintenance dialysis. As reported in a Finnish study, indeed, the costs of a kidney transplant patient for the healthcare system after the first year of transplantation are $<1/3$ of those seen during dialysis treatment; the median annual costs were around 53.000 EUR per patient in dialysis, 59.000 EUR for the first post-transplantation year and around 12.000 EUR for the subsequent years [9]. Diverse local healthcare systems, social insurance and policies among different countries influence an average estimate of costs for both dialysis and transplantation; however, DGF increases annual costs compared to patients with EGF, causing economic repercussions on healthcare system [9]. Moreover, without significant improvement of graft function within 6 to 14 days after transplantation, a graft biopsy is required, influencing further post-transplant management costs. Finally, besides economic commitment to healthcare system, DGF has also social and psychological implications as the patient is often overwhelmed due to the frequency of clinic visits and being away from home [10].

2. Risk factors for DGF onset

Multiple risk factors are implicated in the onset of DGF after kidney transplantation, and they are classified as follows: donor-related factors [*i.e.* duration of ischemia, age, Kidney Donor Profile Index (KDPI)], recipient-related factors (immunological response, ischemia-reperfusion injury, dialysis duration), and factors related to surgical procedures. Recently, in particular, DGF incidence is increased because of the use of donor kidneys from Expanded Criteria Donors (ECD) and after Cardiac Death (DCD); indeed, even though DCD kidney transplants have proved to be useful to overcome the shortage of organs, the literature reports an increased risk for Primary Non Function (PNF) and DGF [10–12]. DGF has been described to occur in more than 50% of ECD transplants (all deceased donors $\geq$ 60 years and donors more than 50 years with at least two conditions among cerebrovascular cause of death, history of hypertension, serum creatinine $>$ 1.5 mg/dL) compared with those from standard-criteria [13]. Clinically, DGF is considered a manifestation of acute kidney injury (AKI), resulting in post-transplantation oliguria, increased allograft immunogenicity and decreased long-term survival. A cold ischemia time more than 24 h, high donor proteinuria levels, donor terminal serum creatinine, type of dialysis prior to transplantation or sensitization in re-transplanted patients, increase the incidence of DGF [13–15]; also Warm Ischemia Time (WIT) higher than 18 min was considered a risk factor for DGF [13]. In addition, biologically older allografts are unable to sufficiently resolve inflammatory responses due to epigenetic and environmental factors that decrease resilience to stress and impair physiological function leading to post-transplant worse outcomes [16]. Both Ischemia and Reperfusion after hypothermic preservation cause a complex sequence of events that activate various programs of cell death (*e.g.* necrosis, apoptosis, autophagy-associated cell death), leading to renal injury and development of DGF [17]. Ischemia causes decreased oxygen delivery, intracellular acidosis (by accumulation of lactic acid, due to switch to anaerobic glucose metabolism), inhibition of Na/K ATPase pumps with consequent edema and generation of Reactive Oxygen Species (ROS) [18]. In addition, reperfusion causes an increase in oxygen and pH normalization, with complete activation of calcium-dependent proteases (as Calpain, partially activated by increase of intracellular Ca^{2+} during ischemia) [18]; McGuinness and colleagues reported that the magnitude of changes in response to reperfusion was exacerbated in allografts exhibiting DGF compared to Immediate Graft Function (IGF), maybe due to differential capacity to restore physiological homeostasis following IR [16]. Interestingly, authors also reported an increase in DNA methylation within the promoter and intragenic regions of the DGF-associated genes in renal tissues [*e.g.* Corin, Ubiquitin D (UBD), β-tubulin] in pre-perfusion allografts exhibiting IGF compared to DGF, reversed after reperfusion [16]. Moreover, IR is associated with microvascular dysfunction, activation of innate and adaptive immune system, endothelial-mesenchymal transition, coagulation and Complement cascade. The activation of Complement occurring in IR increases inflammatory response and subsequent renal tissue damage by the release of pro-inflammatory mediators [18–20]. In addition, Complement may be involved in the maladaptive repair leading to the progression to renal fibrosis and CKD; thus, Complement inhibitors [*e.g.* eculizumab, C1-INH, engineered forms of Complement receptor type 1 (CR1) or synthetic inhibitors of Complement convertases (Compstatin)], preventing AKI onset and CKD progression, might be used to improve graft function [21]. Interestingly, in the settings of DGF patients, Complement factors and thrombin synergistically induce a Th1 immune response, which has been suggested to play a key role in the development of transplant rejection; in DGF graft biopsies, in particular, infiltrating myeloid dendritic cells (DCs) expressed Protease-activated Receptor 1 (PAR-1), the main thrombin receptor, whose activation may induce Complement production and Th1 bias [22]. Considering the wide spectrum of mechanisms involved in DGF, the identification of early biomarkers of DGF to implement conventional clinical practice and improve allograft management is necessary [15,23]. Even though DGF onset has multifactorial etiology, new several polymorphisms in genes involved in oxidative stress, telomere length, cytokine secretion, immunity and inflammation have been associated with DGF, allowing us to assess the increased risk of early renal dysfunction after kidney transplantation [24,25]. Single Nucleotide Polymorphisms (SNPs) in Tumor Necrosis Factor (TNF) and Multidrug Resistance-related Protein-2 (MRP-2) genes would indeed infer a genetic predisposition to DGF (Fig. 1) [5]. In addition, the rs7851696 polymorphism in Ficolin-2 gene, an activator of the Complement system *via* the lectin pathway, was associated with an increased risk of DGF and acute kidney allograft rejection in deceased-donor renal transplant recipients [26].

3. Limits of conventional biomarkers

Current conventional methods for the evaluation of renal allograft, such as serum creatinine measurement, Glomerular Filtration Rate (GFR), proteinuria and needle biopsy, are usually ineffective to address the disease progression risk, to predict the clinical outcome or the guide nephrologist in the best treatment option. Moreover, renal needle biopsy is a highly invasive procedure. Serum creatinine is the current "gold standard" to assess kidney function and estimate Glomerular Filtration Rate (GFR), but it is a late and nonspecific marker of function as many factors (diet, muscle mass, metabolism, drugs or hydration status) may affect its reliability [15,27,28]. A renal biopsy is a useful tool to monitor graft function and differentiate ATN, main cause of DGF in kidney transplant recipients, from other causes of graft dysfunction (*e.g.* acute rejection); however, it is an invasive procedure, requiring hospitalization, and it presents sampling error and subjective variability between pathologists [15,28].

4. Emerging biomarkers of DGF

In the latest years, the use of high-throughput omics technologies has led to the discovery of many candidate biomarkers of renal injury. The major goal of these technologies is the early prediction of DGF before

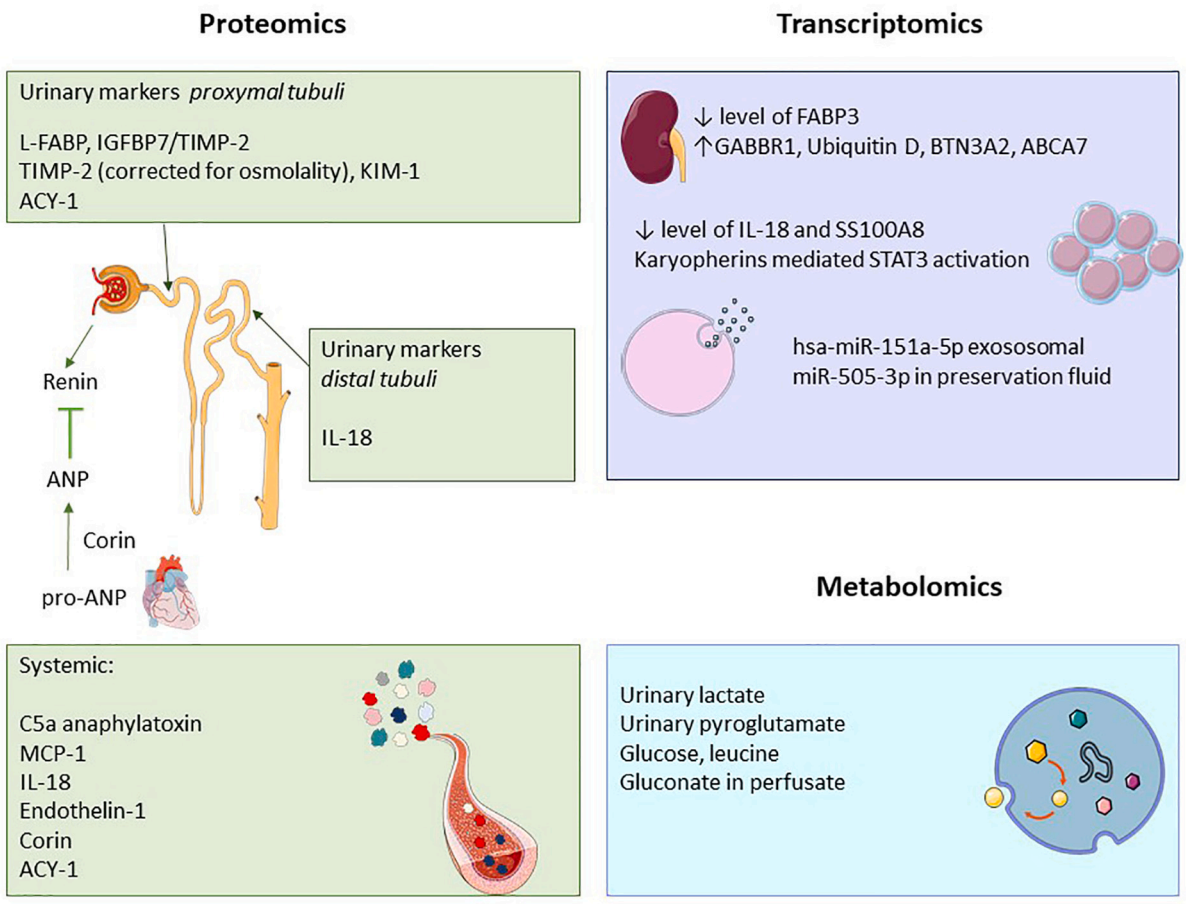


Fig. 1. Emerging biomarkers of DGF.
A wide range of noninvasive biomarkers, discovered by Omics technologies, have been proposed for early identification and stratification of DGF in renal transplantation.

and immediately after kidney transplantation and to provide a deep understanding the complex mechanisms associated with graft dysfunction (Table 1). Omics approaches are high throughput experiments that allow to obtain, in a single experiment, a large amount of data about a specific type of molecule, such as the three billion base pairs of the human genome, the large number of proteins from the renal tissue or a large plethora of metabolites. Examples of these technologies are Next Generation Sequencing (NGS), used for genomics and transcriptomics studies, and mass spectrometry, used in proteomics and metabolomics analysis. However, despite the promising advancement of the recent years, few of these omics biomarkers have been successfully translated to the clinic and are currently used in the choose of the best treatment option.

4.1. Metabolomics and proteomics

Omics techniques lead to the identification of different biomarkers in renal transplantation (*e.g.* Neutrophil Gelatinase Associated Lipocalin [NGAL], Liver-type Fatty Acid Binding Protein 1 [L-FABP], Interleukin18 [IL-18]; Fig. 1) but unfortunately their clinical application is still limited; in response to tubular injury, indeed, these proteins are upregulated and secreted in urine but they have often low specificity, as they increase also in patients with urinary tract infections and sepsis [15,29]. Recent clinical studies have suggested the use of multi-omics data to identify specific metabolic footprints of DGF, combining proteomics and metabolomics approaches, even though bioanalytical metabolomics assays are more challenging due to higher complex physicochemical properties of metabolites compared to proteins [30]; moreover, unlike the genome and proteome, the metabolome is an open system, constantly in flux, that measures the downstream products of multiple genes and proteins and their interactions with food and environment [30]. In renal transplantation, in particular, Lindeman and colleagues reported an impaired oxidative phosphorylation and a compensatory normoxic glycolysis unable to sustain adequate ATP regeneration during the reperfusion phase of the renal graft; they observed, by Magic angle Nuclear Magnetic Resonance (NMR) on tissue and Mass Spectrometry on plasma, altered pathways in DGF grafts compared to EGF grafts: post-reperfusion metabolic incompetence (*e.g.* impaired recovery of the high-energy phosphate-buffer phosphocreatine and persistent ATP/GTP catabolism with consequent continued release of hypoxanthine and xanthine), post-reperfusion autophagy (*e.g.* exclusive release of serine, methionine, tyrosine along with disposal of butyryl- and isovalerylcarnitine and deamination products of branched chain amino acids) and cell damage in a context of mitochondrial failure (*e.g.* release of uracil, an established marker of cell injury) [31]. Interestingly, Lindeman and colleagues speculated that this general metabolic collapse may interfere with any pharmaceutical intervention that relies on the availability of ATP [31]. By an integrative proteo-metabolic study in rat kidney tissue exposed to IRI, also Huang and colleagues reported evident metabolic changes already after 4 h of reperfusion, with elevated levels of metabolites of glycolysis, lipids and Fatty Acids (FA) and impaired ATP production after 24 h; strikingly, using a non-lethal unilateral long IRI model in the rat, they noted significant metabolic effects in contralateral and separate healthy kidneys used as control, detecting higher levels of urea and AMP after 24 h; they speculated that this metabolic behavior in healthy kidney could be a partial

Table 1
DGF biomarkers and their function.

Markers	References	Study design	Biological sample	Function in DGF
Proteomics				
TIMP-2	Bank et al. [43]	Study on 74 DCD recipients and 51 of them (69%) experienced fDGF; samples collected on postoperative days 1 to 7, day 10, week 6, and month 6, and values corrected for osmolality (mOsm)	Urine	Increase
C5a anaphylatoxin	Schroppel et al. [44]	Prospective cohort study on 469 deceased donors and their corresponding 902 kidney recipients; DGF in the cohort was 32% (n = 291)	Donor urine	Increase
Klotho	Castellano et al. [20]	A study on animal and human models; 7 patients experienced DGF and 7 were controls; down-regulation of tubular Klotho expression with deficiency of soluble Klotho after transplantation	Serum	Decrease
Endothelin-1	Perico et al. [1]	Review discussing experimental studies on human and rat models about ischemia-reperfusion injury and DGF	Serum	Increase
Corin	Hu et al. [45]	27 deceased donor transplantations: 11 experienced DGF, 16 IGF; sample collected at 24 h post operation	Plasma	Decrease
Aminoacylase-1 (ACY-1)	Welberry Smith [46]	Mass spectrometry analysis before and on day 2 post transplant from 5 patients with DGF and 5 with an uncomplicated transplant for proteomic discovery; in 2 validation cohorts showed as long-term outcome biomarker	Serum	Decrease
Transcriptomics				
UBD mRNA	McGuinness et al. [16]	RNAseq cohort: 11 DGF patients, 12 IGF; DGF-specific targets adjusted for donor gender	Renal tissue biopsies	Increase
Karyopherin	Zaza et al. [50]	A study analyzing the transcriptomic profile of PBMC from 24 renal transplant recipients with DGF and with EGF, before (T0) and 24 h (T24) after transplantation	Recipient PBMC	Increase
Exosomal miRNAs (hsa-miR-151a-5p)	Wang et al. [52]	miRNA expression profiling of exosomes in the peripheral blood from 9 DCD recipients	Recipient peripheral blood	Increase
miR-505-3p	Roest et al. [53]	48 kidney grafts from DCD or DBD donors; miRNA profiles from vascular bed preservation fluid	Graft preservation fluid	Increase
Metabolomics				
Lactate	Kostidis et al. [29]	Randomized controlled trial on 92 DCD kidney transplant recipients; urine samples obtained 10 days after transplantation, at days 42, 180, and 365	Urine	Increase
Pyroglutamate	Kostidis et al. [29]	Randomized controlled trial on 92 DCD kidney transplant recipients; urine samples obtained 10 days after transplantation, at days 42, 180, and 365	Urine	Decrease
Lactate/fumarate ratio	Kostidis et al. [29]	Randomized controlled trial on 92 DCD kidney transplant recipients; marker suggested as strong predictor of prolonged fDGF (≥ 21 days)	Urine	Increase
BCAAs/pyroglutamate ratio	Kostidis et al. [29]	Randomized controlled trial on 92 DCD kidney transplant recipients; marker suggested as strong predictor of prolonged fDGF (≥ 21 days)	Urine	Increase
Glucose metabolite	Guy et al. [33]	Perfusate of 26 transplanted cadaveric kidneys was analyzed: 19(73%) with IGF and 7(27%) with DGF; perfusates were sampled at 45 min and 4 h of preservation	Perfusate fluid	Decrease
Leucine amino acid	Guy et al. [33]	Perfusate of 26 transplanted cadaveric kidneys was analyzed: 19(73%) with IGF and 7(27%) with DGF; perfusates were sampled at 45 min and 4 h of preservation	Perfusate fluid	Increase
Gluconate metabolite	Guy et al. [33]	Perfusate of 26 transplanted cadaveric kidneys was analyzed: 19(73%) with IGF and 7(27%) with DGF; perfusates were sampled at 45 min and 4 h of preservation	Perfusate fluid	Decrease

Studies investigating biomarkers of DGF in different biological samples in kidney transplantation.

compensation mechanism caused by the acute energy deficit after the ischemic insult in injured kidney [32]. Also the perfusate from human donor kidneys awaiting transplantation is a noninvasive biological fluid useful to understand the metabolomic processes occurring in preserved organs; Guy et al. reported a significant metabolic activity during Hypothermic Machine Perfusion (HMP) and identified, through NMR spectroscopy, different metabolomic profiles for DGF and IGF kidneys that could allow to predict functional graft outcome after transplantation [33]. In particular, they observed that glucose concentrations were significantly lower in the DGF kidney compared to those with IGF at 45 min and 4 h of perfusion, suggesting that more glucose is required for repair by damaged DGF kidneys after ischemia-reperfusion injury (Fig. 1); moreover, they reported raised concentrations of amino acids, as leucine, in the perfusate of DGF kidneys at 45 min, suggesting an increased cellular breakdown in damaged organs (Fig. 1). Conversely, gluconate, one of the constituents of the Kidney Perfusion Solution (KPS-1), showed unexpectedly reduced concentrations at 4 h, probably due to influx of gluconate into the damaged cells of DGF kidney (Fig. 1) [33]. In addition, Kostidis and colleagues analyzed urinary metabolome of Donors after Circulatory Death (DCD) transplant recipients at multiple time points (days 10, 42, 180 and 360 after transplantation) and showed that, in prolonged fDGF (≥21 days), urine lactate was significantly higher and pyroglutamate lower than in patients with limited fDGF (Fig. 1), suggesting the altered urinary metabolome as a marker to predict fDGF [29]; furthermore, they reported that the significant increase of two ratios of urinary metabolites, lactate/fumarate and Branched-ChainAmino Acids (BCAAs)/pyroglutamate, predicted prolonged fDGF [29]. Interestingly, urine proteomic signatures in the donor could predict early post-transplant function; Field and colleagues suggested, indeed, urinary KIM-1 as a potential marker of AKI in deceased cardiac organ donors as this biomarker was increased in donors whose kidneys displayed aberrant early function, but it is useless in the case of DGF [15,34]. Also, donor urinary marker MCP-1, a chemokine produced by many cell types as endothelial, fibroblasts and monocytes cells and involved in inflammation and repair after kidney injury, has limited clinical utility in prediction of subsequent recipient graft function [35,36] or incidence of acute renal allograft rejection [25,36]. However, de Vries and colleagues demonstrated that certain cellular infiltrates (*i.e.* T lymphocytes and macrophages), contained in kidneys of Brain Dead Donors (DBD) before transplantation, may enhance the proinflammatory state of the organ releasing mediators upon reperfusion; these released chemokines as MCP-1, in turn, may attract recipient leukocytes to the kidney and, consequently, increase the risk of acute rejection [37]. Moreover, Prodjosudjadi et al. hypothesized a local allograft production, mainly by tubular epithelial cells but also by monocytes and macrophages, as they observed absence of significant changes in serum levels of MCP-1 but an increase of its urinary excretion during acute renal allograft rejection, suggesting that urinary MCP-1 may be a marker of acute rejection [38]. In a study conducted on patients with acute renal graft dysfunction, Grandaliano et al. reported

that an increased MCP-1 urine excretion correlated with increased MCP-1 gene and protein expression in renal biopsies, mainly in proximal tubular cells, and the MCP-1 expression was significantly higher in patients with AR than patients with a specific Acute Tubular Damage (ATD) [39]. Furthermore, Raza et al. reported that the increased excretory levels of MCP-1, quantified by MCP1/CCL2 ELISA test using urine samples collected prior biopsy procedure, were associated with biopsy-proven rejection episodes and allowed to distinguish a severe form of T cell-mediated rejection [*i.e.* Acute Vascular Rejection (AVR)] from mild rejection type [Bordenline (BLR)] [40]. Interestingly, the cell-cycle arrest biomarkers Tissue Inhibitor of MetalloProteinases 2 (TIMP-2) and urinary Insulin-like Growth Factor-Binding Protein 7 (IGFBP7) have been recently discovered and validated as predictive biomarkers of AKI and their trajectories predicted short- and long-term outcomes [41,42]; few studies analyzed these biomarkers in the transplant setting. Bank and colleagues observed comparable or even better performance of urinary 24-h creatinine excretion to urinary TIMP-2 corrected for osmolality (TIMP-2/mOsm; Fig. 1); these authors observed higher TIMP-2 levels in the urine of DGF patients and, interestingly, this marker followed a "trajectory" in any patients because it was higher on day 10 in prolonged DGF and its decrease preceded the increase of GFR and the resolution of DGF [11,43]. Finally, Schroppel and colleagues found that elevated levels of the C5a anaphylatoxin, released at the time of organ procurement in donor urine without AKI, were associated with an elevated risk of DGF (Fig. 1) [44]. In addition to urinary biomarkers, serum proteins were associated with DGF; Castellano and colleagues reported that level of Klotho were reduced in the serum of DGF patients and, considering pleiotropic functions as inhibition of fibrosis and preservation of endothelial function, they hypothesized that Klotho deficiency may play a significant role in DGF-associated chronic allograft dysfunction [20]. Noteworthy, in a swine model, they also found a profound down-regulation of tubular anti-senescence and anti-fibrotic protein Klotho in proximal Tubular Epithelial Cells (TEC) after C5a anaphylatoxin stimulation and NF-kB signaling activation, and a statistically significant reduction of tissue Klotho expression in human allograft biopsies from DGF patients compared to corresponding pre-implantation cadaveric kidney donors, confirming the pathogenic contribution of the Complement cascade to IR injury and DGF [20]. Moreover, Perico et al. reported that patients with DGF had high serum concentration of the potent vasoconstrictor Endothelin-1, which might perpetuate post-transplantation renal damage (Fig. 1) [1]. Furthermore, Hu et al. suggested plasma Corin as potential biomarker of DGF as they observed reduced levels in recipients presenting DGF after IR injury; Corin, a serine protease that cleaves the precursor and generates the potent endocrine factor Atrial Natriuretic Peptide (ANP), protects against kidney injury and it may be a potential therapeutic candidate for AKI in kidney transplantation (Fig. 1) [45].

Finally, comparing pre- and postoperative serum proteins from patients with and without DGF using immunodepletion followed by label-free single dimensional liquid chromatography-tandem mass spectrometry analysis, Welberry Smith and colleagues suggested aminoacylase-1 (ACY-1; a zinc-binding homodimeric cytoplasmic protein highly expressed in kidneys) as a useful biomarker to stratify DGF subgroups in terms of long-term outcomes, especially in the DBD transplant. They reported ACY-1 as a marker of repair/response to damage, as DBD donor transplant recipients with lower concentrations immediately post-transplant showed poorer outcomes [46]; interestingly, the involvement of ACY-1 in protection/repair mechanism was confirmed in mouse models by its interaction with the repair protein sphingosine-1-phosphate [46].

Furthermore, in the medical histological field, a promising tool of increasing interest is the Mass Spectrometry Imaging (MSI), which allows to identify molecular patterns of early kidney injury and to correlate the localized chemical profiles with tissue and morphology; van Smaalen and colleagues identified indeed, in a porcine donor-kidney model, surrogate markers of ischemic processes in renal tissue. In particular, they reported significantly elevated lysophospholipids in severe ischemia compared with mild ischemia, maybe due to lipid degradation by increased activation of phospholipase A1 (PLA1) or PLA2; these authors suggested MSI as a practical tool to identify molecular patterns of early ischemic injury in a clinically acceptable time frame and to predict transplant-outcome [47].

4.2. Transcriptomics

The analysis of expression of many genes involved in oxidative stress and tissue repair could help to discover new emerging biomarkers, as the reduced levels of Fatty Acid Binding Protein (FABP3) in pre-implantation biopsies that were associated with DGF (Fig. 1) [48]. Moreover, reduced levels of IL-18 and S100A8 transcripts in peripheral mononuclear cells, measured by quantitative RT-PCR, predicted immediate and DGF (Fig. 1) [49]. Interestingly, studying the correlation between allograft biological age and adverse outcomes post-transplantation as DGF, McGuinness et al. determined a molecular signature for impaired physiological function, *via* a synchronous genome, transcriptome and proteome snapshot using human renal allografts as an *in vivo* model of aging in humans [16]. Ingenuity Pathway Analysis (IPA) indicated the activation of innate immune responses, the cell death/survival, the cellular fitness and the cell-cell communication as top-ranked networks during DGF occurrence; the transcriptome biotype observed only in DGF, but not in IGF, included an increase in antisense, pseudogenes, coding and non-coding RNA, consistent with allostatic load. By differential gene expression between two experimental groups (pre- *versus* post-perfusion biopsies, and DGF *versus* IGF biopsies), six transcripts were found to be significantly different between DGF and IGF after adjustment for donor gender: Gamma-AminoButyric Acid Type B Receptor subunit 1 (GABBR1; Fig. 1), Ubiquitin D (UBD; Fig. 1), Butyrophilin subfamily 3 member A2 (BTN3A2), ATP Binding Cassette subfamily A member 7 (ABCA7), Regenerating Family Member 1 Beta (REG1B) and Deleted in Azoospermia 1 (DAZ1). In particular, GABBR1 has been associated with proteasome activation, while UBD has been implicated in inflammatory-mediated signaling, directing its substrates for 26S proteasomal degradation and autophagy during nutrient-deprived conditions, probably supported by significant transcriptome changes in mitochondrial energy metabolism (*e.g.* HSP90AA1, HSPA1A, HSPA1B transcripts) [16]. Three of them (GABBR1, UBD, BTN3A2) locate close to the major Histocompatibility class 1 (HLA) locus, suggesting a potential involvement of this HLA locus in DGF occurrence [16]. As targets for pharmacological interventions in renal transplantation, Zaza and colleagues investigated the transcriptomic profile of PBMC in transplant recipients with DGF and EGF and suggested that Karyopherins (Fig. 1), a group of nucleocytoplasmic transporters, reproductive biomarkers of clinical DGF. Karyopherin-mediated shuttling of intracellular molecules (*e.g.* transcription factors), indeed, may be involved in the onset of DGF in renal transplant; *in vitro* functional analysis showed significant karyopherin-mediated translocation of phosphorylated STAT3 in PBMC of DGF patients [50].

Moreover, microRNAs are emerging biomarkers in kidney injury and transplantation; they are small noncoding RNAs of approximately 22 nucleotides that regulate the expression of hundreds of mRNAs. McGuinness and colleagues reported that microRNA profiles in pre-transplant renal biopsies, involved especially in senescence pathways of ischemic injury, are a potential marker of post-transplant allograft function [51]. A better understanding of how miRNAs affect gene expression in post-transplant injury is necessary before integrating novel therapeutics in clinical practice [17]. By high-throughput sequencing, in particular, upregulated miRNA expression profiling of exosomes has been analyzed in the peripheral blood of kidney recipient with DGF. Exosomal miRNAs (*e.g.* hsa-miR-151a-5p), targeting putative genes mainly involved in MAPK signaling pathway is known to play a pivotal role in chronic allograft dysfunction, could be candidate biomarkers for DGF (Fig. 1); consequently, their inhibitors might offer a novel

treatment in renal transplant [52]. Moreover, using conventional reverse transcription and quantitative PCR detection, Roest et al. identified miR-505-3p in preservation fluid from DCD grafts as an independent predicting factor for DGF (Fig. 1); they suggested that increased levels of miR-505-3p is a result of its release from endothelial cells in response to this vascular injury [53]. In addition, a urine microRNA signature is suggested as a noninvasive measure of DGF risk in kidney transplant recipients (Fig. 1) [54].

5. Conclusion

Novel potential DGF biomarkers have been discovered by omics technologies; however they need further validation and require more insights. Indeed, the translation of biomarkers from discovery to clinical practice is still hard, due to the lack of universal procedures for the study design (*i.e.* handling and processing of biological samples), lack of standard guidance for method validation (*i.e.* evaluation of sensitivity, specificity, reliability of the analytical tests) and absence of adequate prospective and retrospective cohort studies to allow inter-laboratories reproducibility in a large population. Consequently, evaluating also the cost-utility and cost-effectiveness of new clinical tests, there is limited success to translate *benchwork* to clinical application, resulting in a small number of patent applications and low number of successful market applications. Besides bureaucratic requirements to new biomarker validation, to date there is no single molecular marker that reflects the histopathological conditions of the kidney in all its complexity, unlike serum creatinine; therefore, the combination of a panel of different biomarkers may help clinicians to identify early graft dysfunction, reducing the number of grafts biopsies and increasing the development of new effective therapeutics. On the basis of the above, the discovery of new non-invasive biomarkers of DGF is critical to improve the clinical management of patients with DGF, implementing the quality of the entire health care system.

Declaration of Competing Interest

The authors declare no conflict of interest.

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